

DATA SCIENCE

CONTINUOUS ASSESSMENT - 4

NAME : TANYA SINGH

PRN : 22070521013

SEC : A

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INTRODUCTION

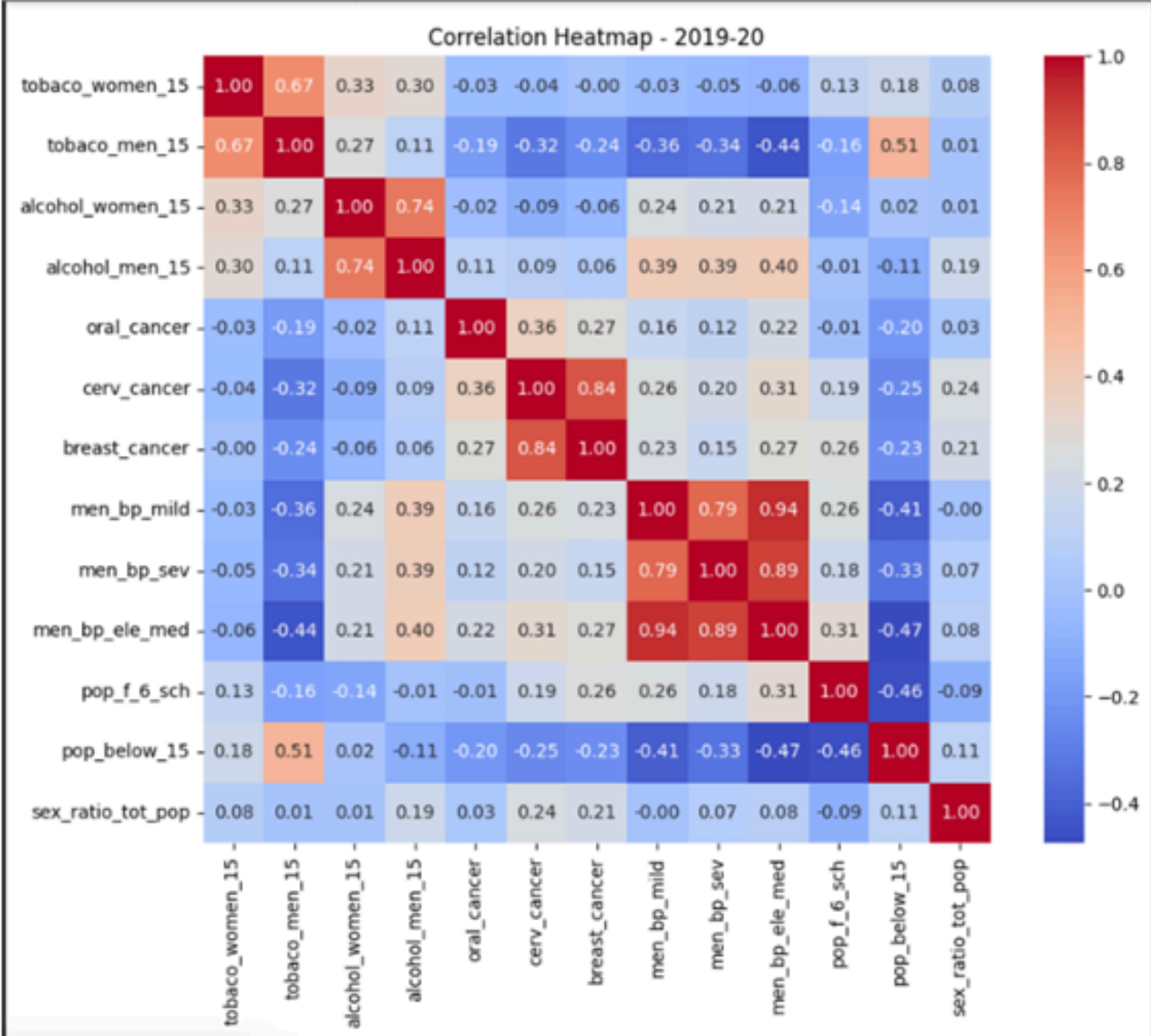
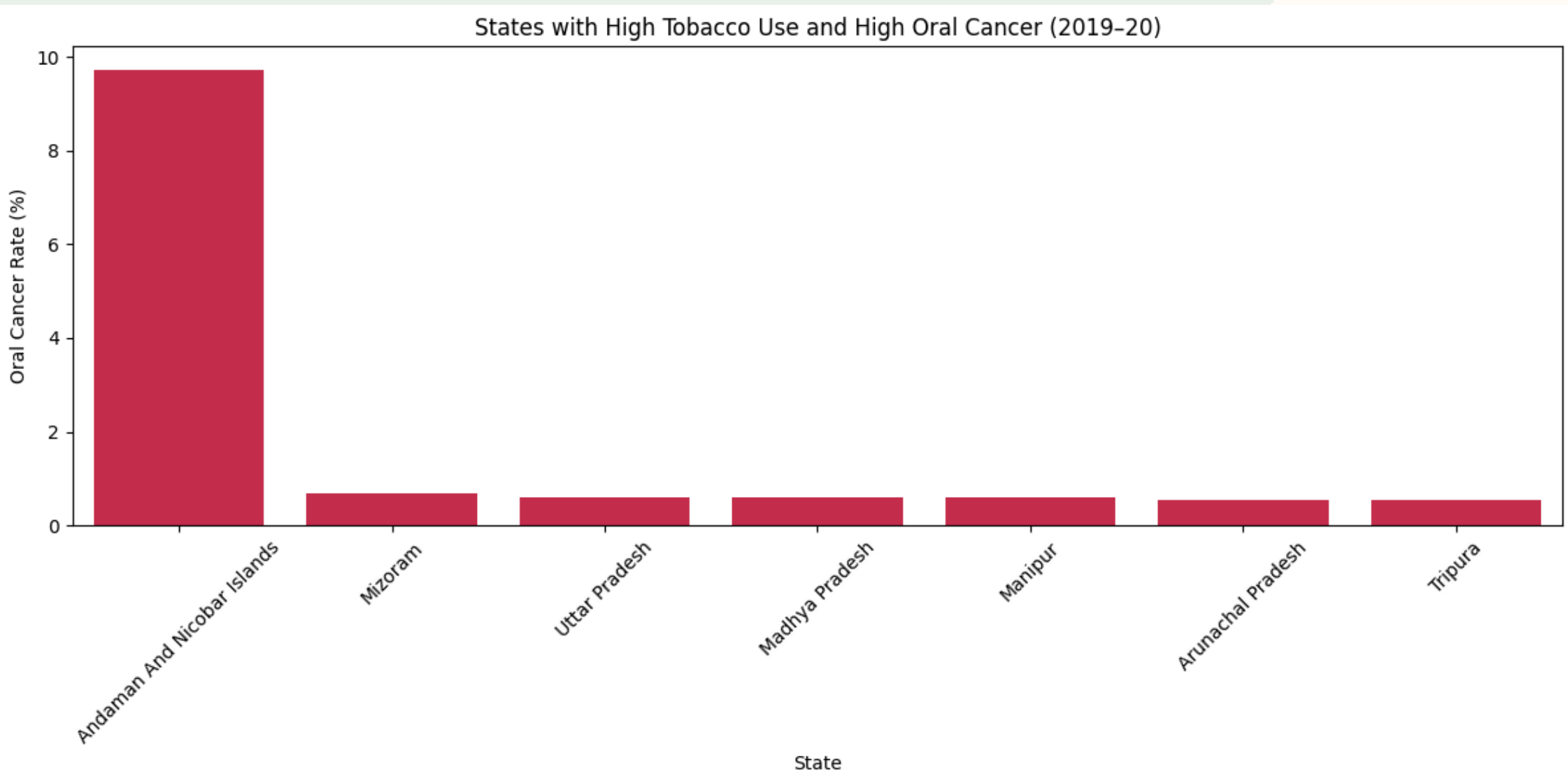
What the Project really is about

- This project analyzes data from the National Family Health Survey from India Data Portal to explore the socio-demographic, behavioral, and health factors influencing teenage pregnancy and women's health outcomes in India.
- Using statistical and machine learning models, including linear regression, clustering, and decision trees, the study identifies patterns linking education, early marriage, anaemia, and nutrition to women's empowerment and reproductive health.
- The dataset covers multiple indicators such as population structure, family planning, child health, and chronic diseases across Indian states and districts.
- I chose this topic because it reflects a real and pressing public-health issue, allowing data science to uncover actionable insights.

DATASET DESCRIPTION

- Source: India Data Portal
- Time Period: July 2014 – March 2015 and July 2019 – March 2020
- Scope: Nationwide – all Indian states and union territories
- Observations: 1,300 district-level records
- Features: 110 variables, with 15 key indicators selected for focused analysis
- Dataset Link : <https://indiadataportal.com/p/national-family-health-survey/r/mohfw-nfhs-dt-qg-eie>

EDA FINDINGS



Analytical Models and Methodology Overview

SIMPLE LINEAR REGRESSION

A simple linear regression was applied to analyze how early marriage (% women married before 18) affects teenage pregnancy (% girls aged 15–19 pregnant) across districts.

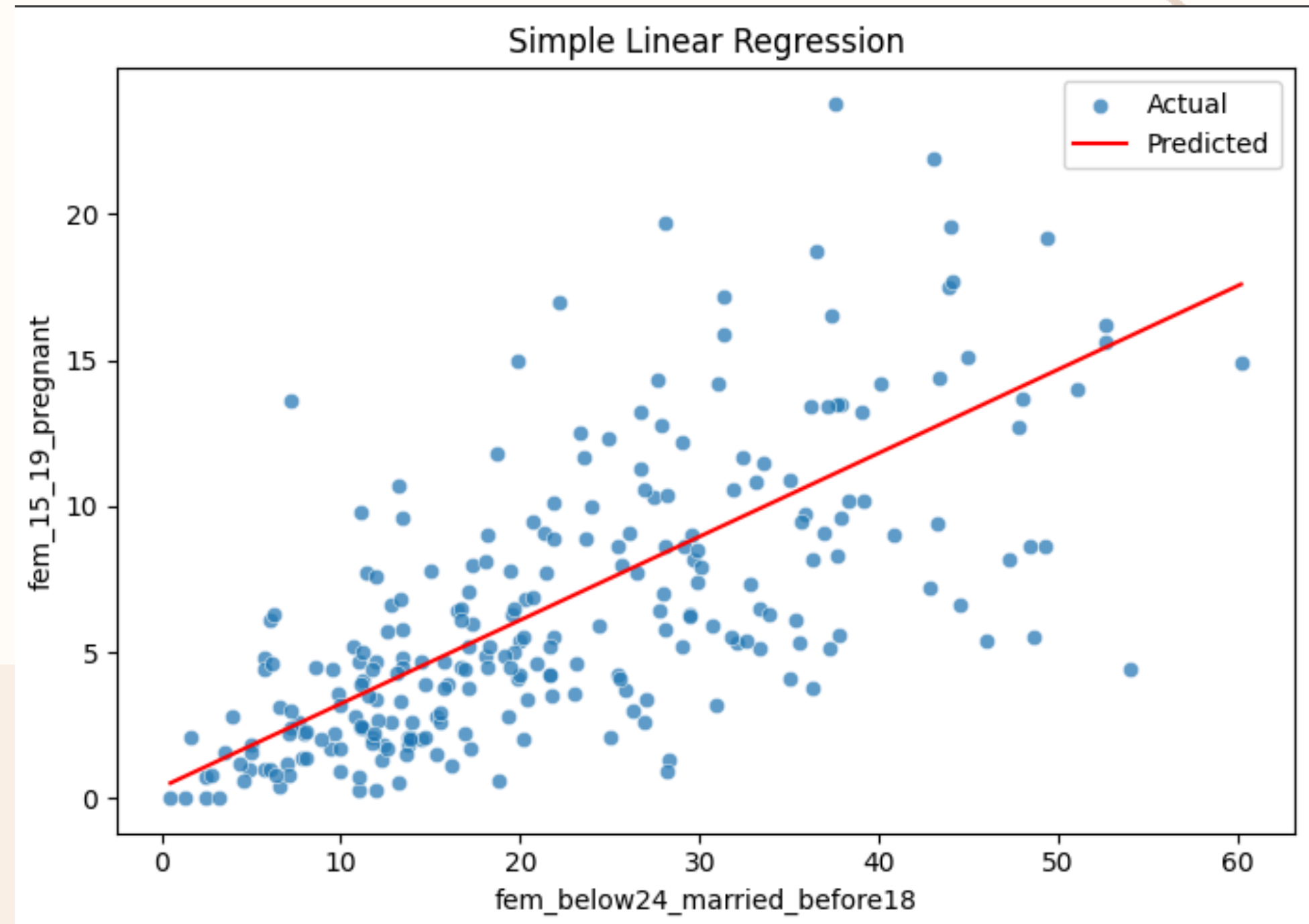
Summary of Results:

Intercept: 0.37 | Slope: 0.29

For every 1% rise in early marriage, teen pregnancy increases by $\approx 0.29\%$.

$R^2 = 0.447$, meaning early marriage explains $\sim 45\%$ of variation in teen pregnancy.

RMSE = 3.5, showing moderate prediction accuracy.



Analytical Models and Methodology Overview

MULTIPLE LINEAR REGRESSION

A multiple linear regression was applied to predict teenage pregnancy (%fem_15_19_pregnant) using five socio-health predictors:

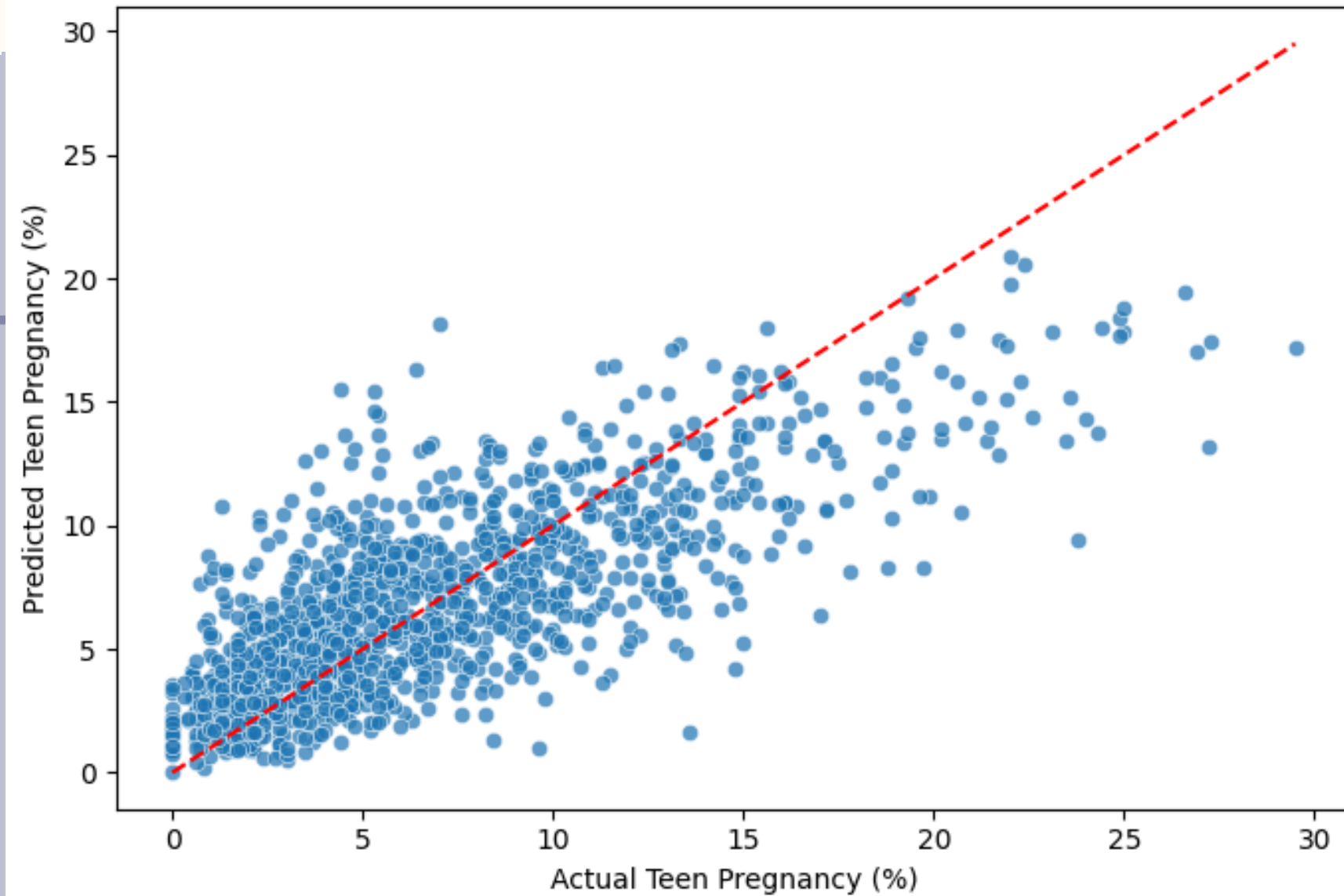
- early marriage rate (fem_below24_married_before18)
- contraceptive use (fp_pill)
- young population percentage (pop_below_15)
- child underweight rate (child_5_underweight)
- anaemia in pregnant women (wom_preg_15_49_anaemic)

Summary of Results:

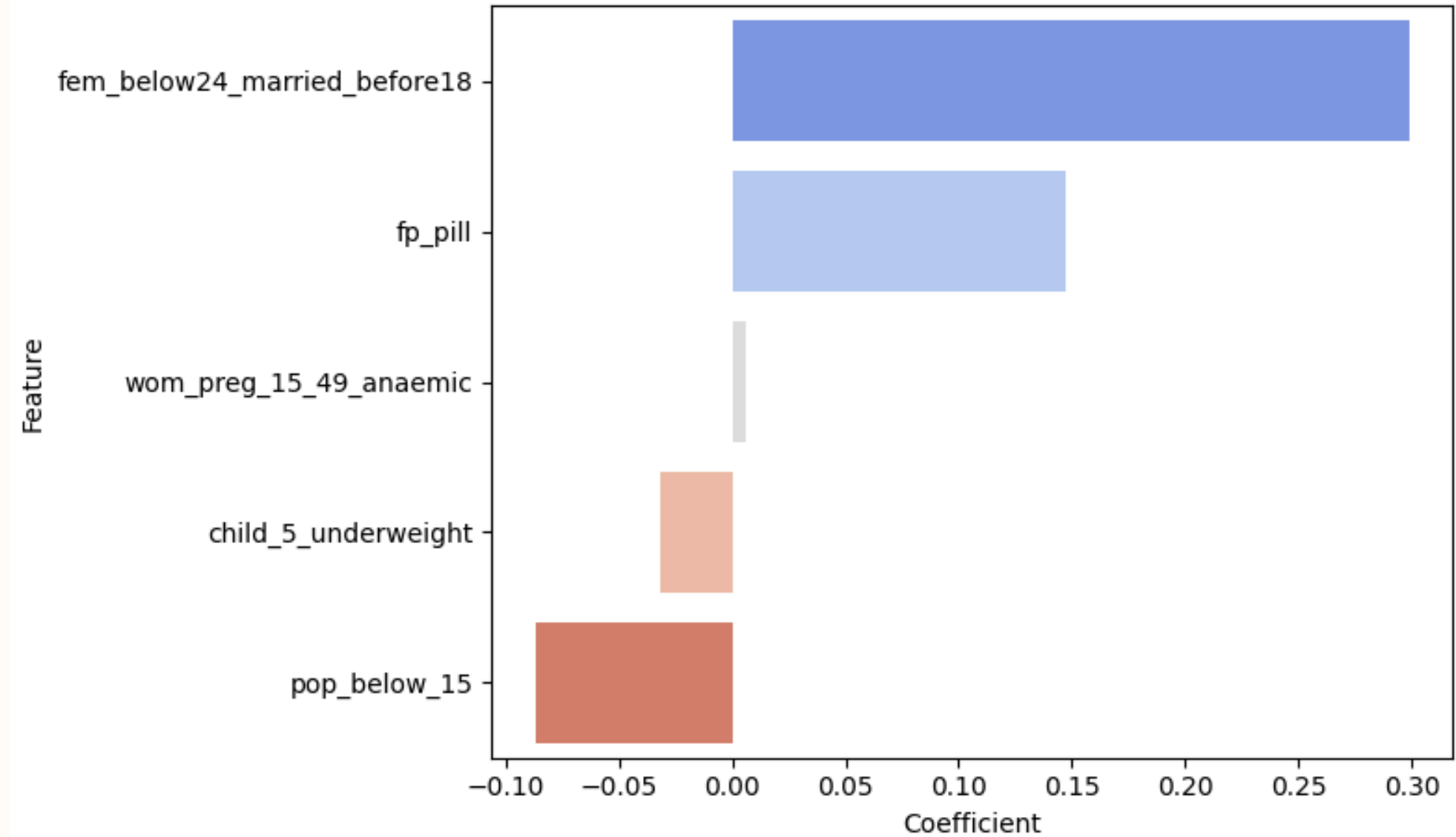
- Model performance improved notably compared to the simple regression:
- R^2 increased from 0.447 $\rightarrow$ 0.587, showing a better overall fit (~30% improvement).
- RMSE decreased from 3.50 $\rightarrow$ 3.30, meaning predictions became more accurate.
- The early marriage variable remained the strongest positive predictor.

Graphs for Multiple Linear Regression

Multiple Linear Regression: Actual vs Predicted



Feature Importance - Multiple Linear Regression Coefficients



Behavioral & Health Risk Clustering (K-Means)

K-Means clustering on behavioral and health indicators formed 6 distinct district groups, revealing strong links between tobacco/alcohol use and disease burden across India.

Purpose:

To group districts based on men's and women's behavioral health indicators.

Variables Used:

- Men's health behaviors: tobacco use (tobaco_men_15), alcohol use (alcohol_men_15), blood pressure severity (men_bp_sev)
- Women's behaviors: tobacco use (tobaco_women_15), alcohol use (alcohol_women_15)
- Disease indicators: oral cancer (oral_cancer), cervical cancer (cerv_cancer)

Cluster 0 – Very Low Risk

Cluster 1 – High Male Tobacco–Alcohol Use

Cluster 2 – Extreme Substance Use Both

Genders

Cluster 3 – Moderate Behavior, High Female Cancer

Cluster 4 – Severe Cancer Burden

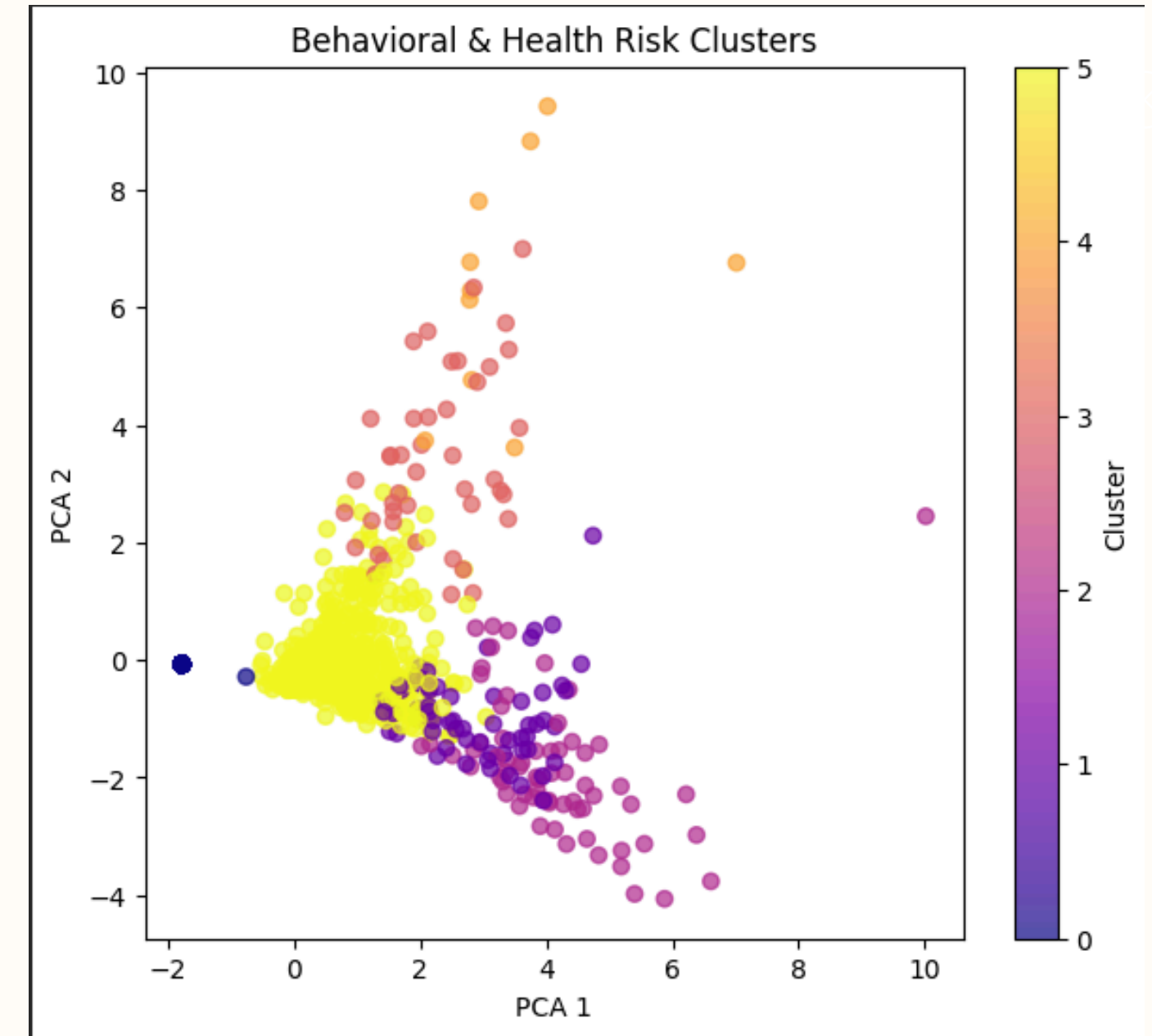
Cluster 5 – Moderate Behavioral Risk

Graphs

```
cluster_centers_b = pd.DataFrame(  
    scaler.inverse_transform(kmeans_b.cluster_centers_),  
    columns=features_behavioral  
)  
print(cluster_centers_b.round(2))
```

	tobacco_men_15	alcohol_men_15	men_bp_sev	tobacco_women_15
0	0.06	0.00	0.00	0.00
1	59.23	27.91	5.84	40.36
2	50.16	49.07	7.75	20.75
3	21.27	30.79	7.67	5.18
4	30.42	26.42	7.23	10.09
5	39.24	18.63	5.71	7.83

	alcohol_women_15	oral_cancer	cerv_cancer
0	0.00	0.00	0.00
1	2.76	0.44	1.50
2	19.60	0.53	0.84
3	1.70	1.80	9.77
4	0.97	10.20	5.29
5	0.95	0.48	0.92



These very low p-values confirm the clusters are statistically distinct — not random groups. It means behavioral patterns and disease outcomes differ significantly across clusters.

```
oral_cancer: F = 545.52, p < 0.001  
cerv_cancer: F = 501.47, p < 0.001  
men_bp_sev:  F = 707.37, p < 0.001
```

Women's Empowerment Clustering (K-Means)

Purpose:

To identify regional patterns in education, early marriage, contraception, anaemia, and teen pregnancy.

Variables:

female schooling, early marriage, teen pregnancy, contraceptive use, anaemia.

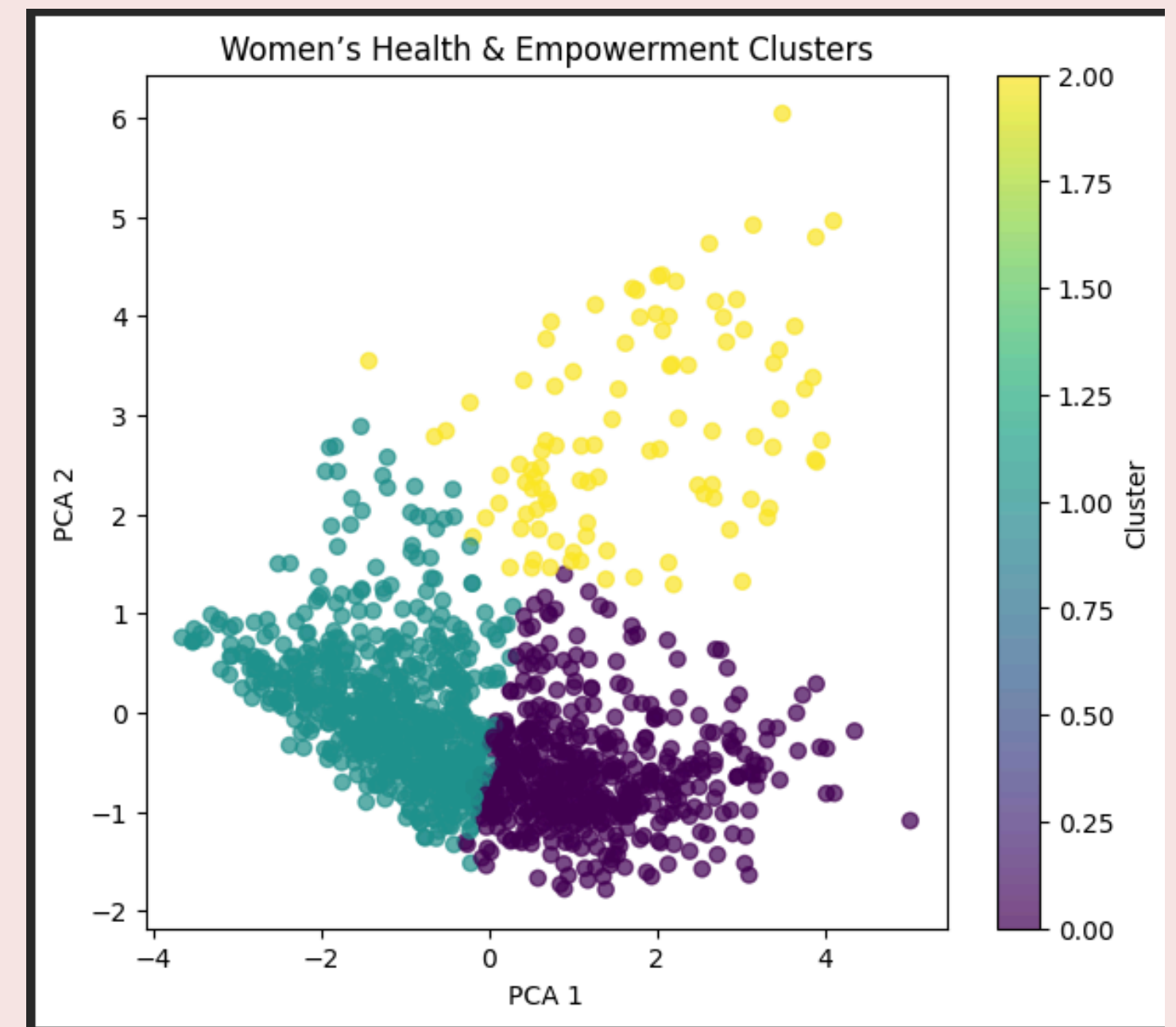
Clusters Found: 3

Findings:

- Cluster 0-Transitional: Mid-level education, high anaemia, low contraception.
- Cluster 1 -Empowered & Healthy: High education, low early marriage, low anaemia.
- Cluster 2 – Educated but At-Risk: High education but high early marriage & teen pregnancy.

Conclusion:

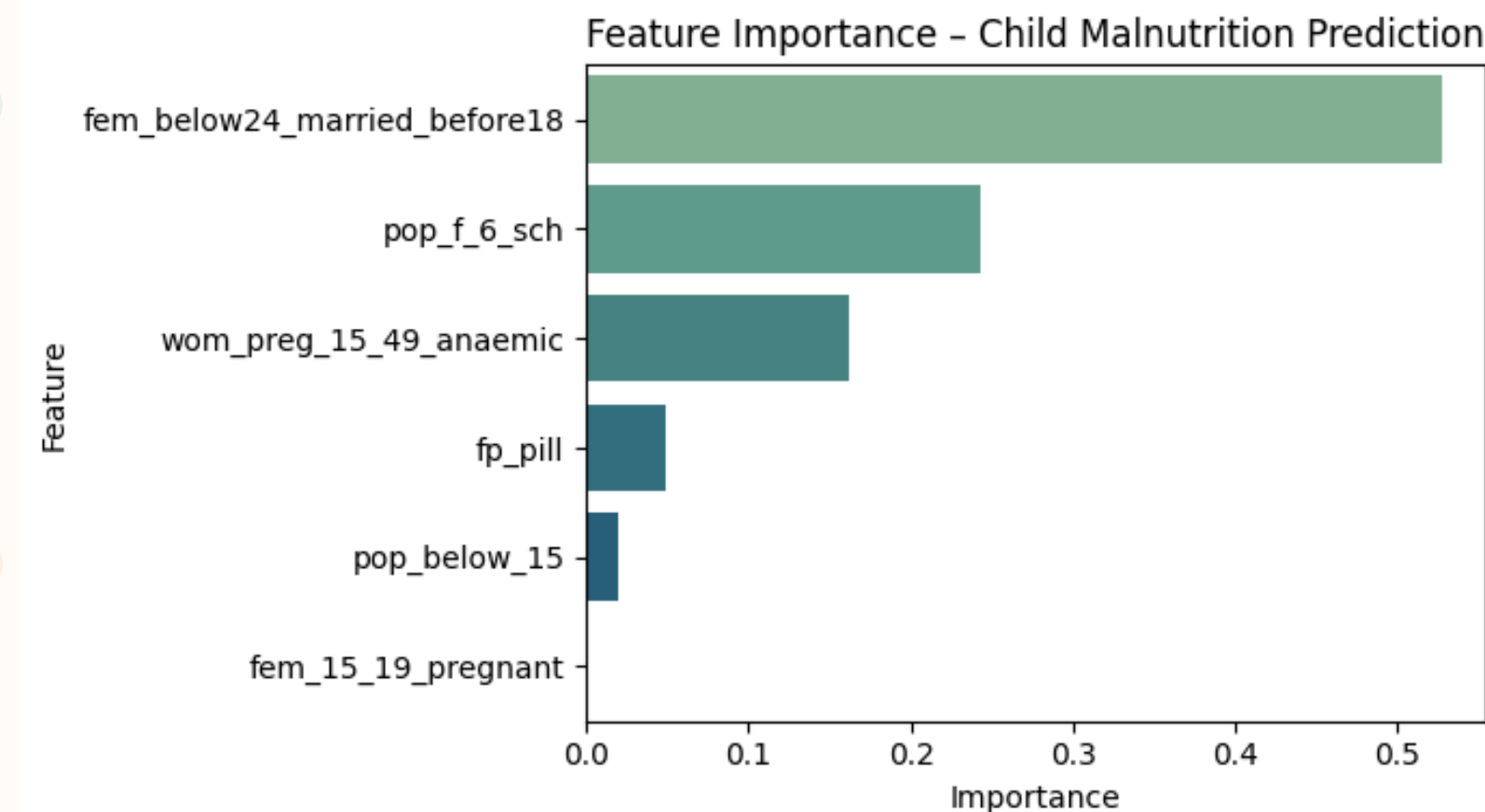
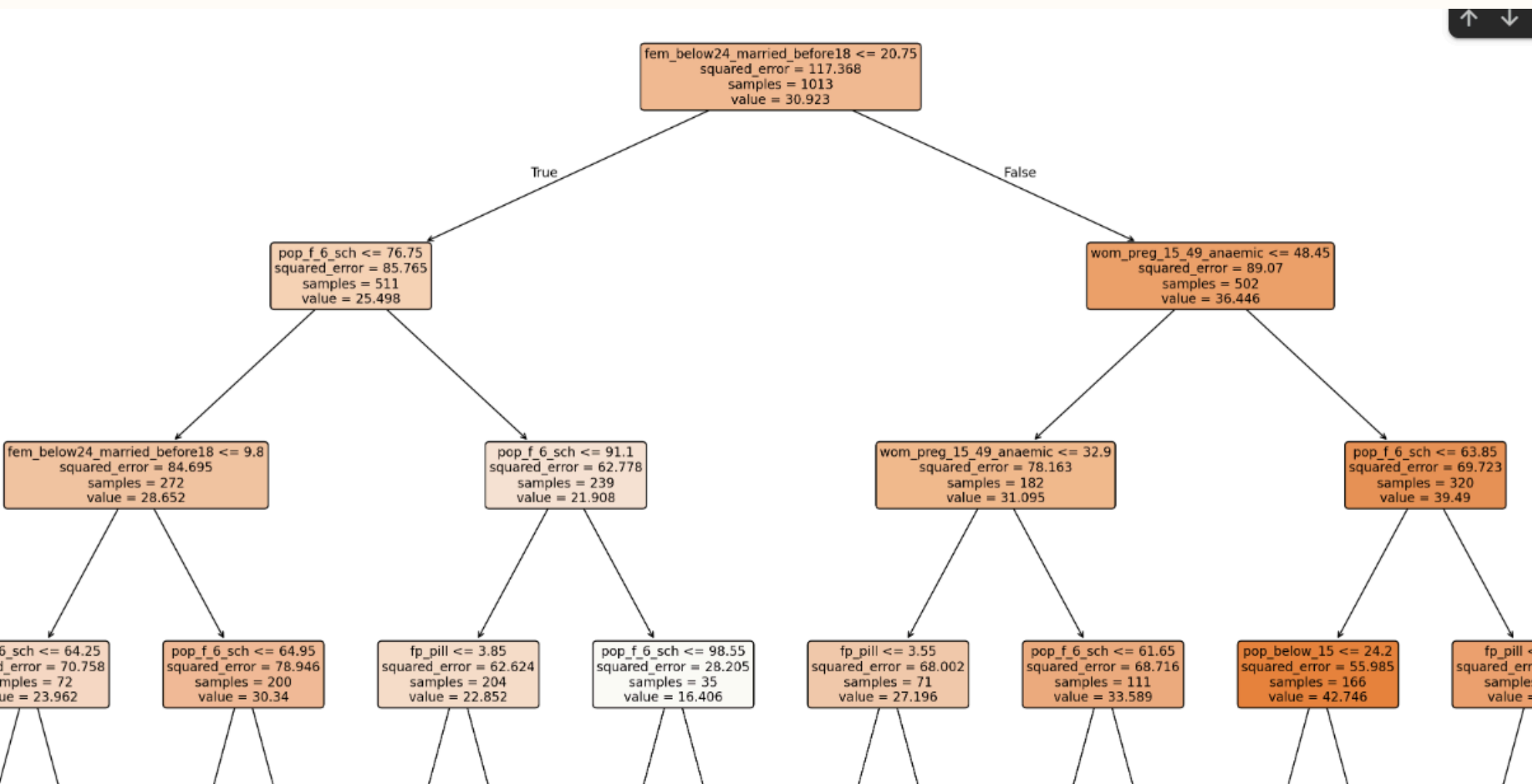
Education alone is not enough true empowerment requires health access and social change together.



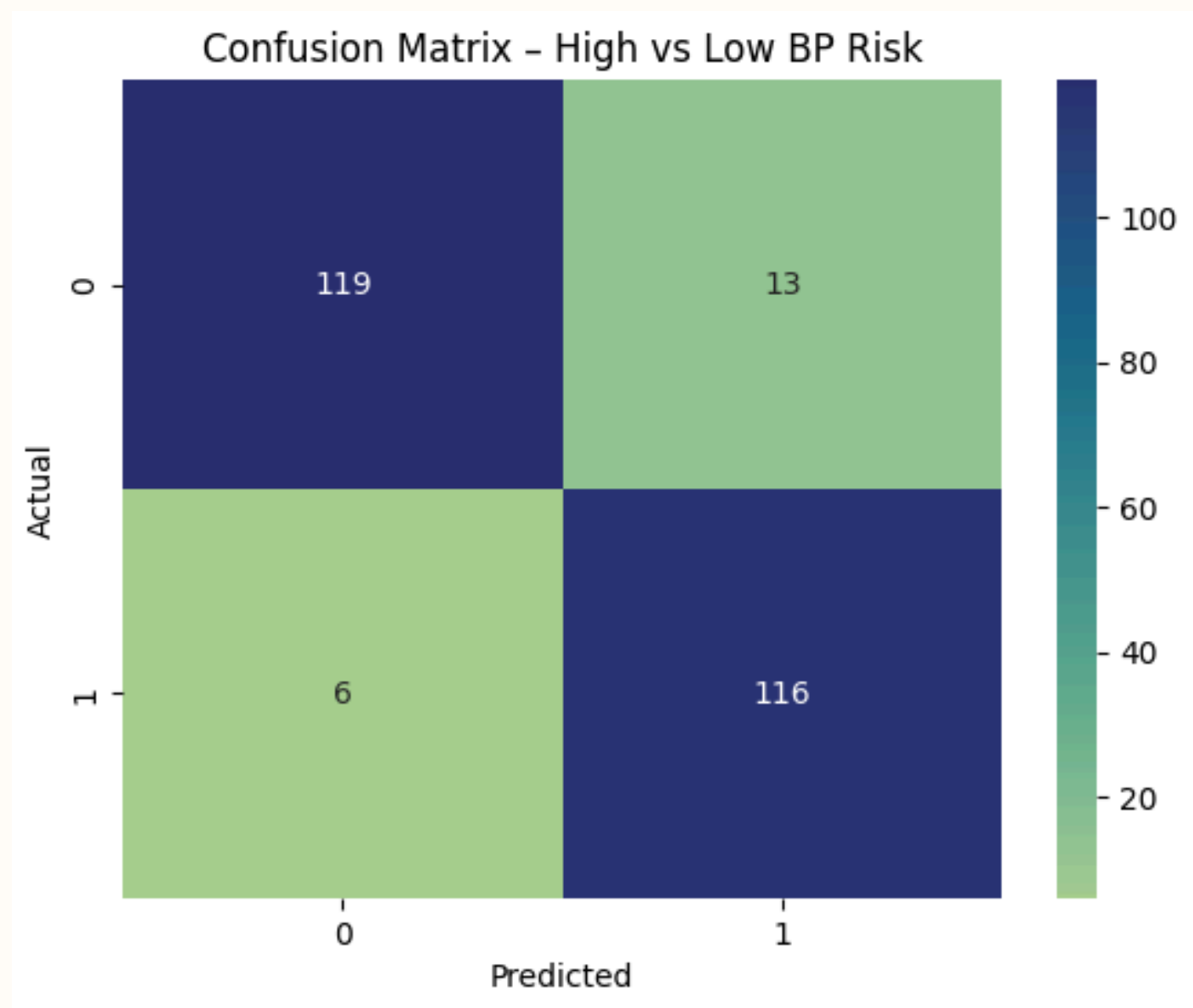
DECISION TREE

To predict the percentage of underweight children under age 5 using a combination of women's health, education, and demographic indicators.

$$R^2 \approx 0.65, \text{RMSE} \approx 3.4$$



LOGISTIC REGRESSION



To classify districts into High-Risk or Low-Risk zones for men's severe blood pressure (BP) using their lifestyle and health factors.

Logistic Regression - Men's High BP Risk				
Accuracy: 0.925				
	precision	recall	f1-score	support
0	0.95	0.90	0.93	132
1	0.90	0.95	0.92	122
accuracy			0.93	254
macro avg	0.93	0.93	0.93	254
weighted avg	0.93	0.93	0.93	254

Model achieved 92.5% accuracy — male tobacco and alcohol use strongly predict hypertension risk among men.

Ground Realities Reflected

6 out of 10

Women aged below 24 were married before 18

1 out of 3

Women (aged 15–49) suffer from anaemia

1 out of 5

Men consume alcohol regularly

3 out of 10

Teenage girls (15–19 years) are pregnant

All statistics are derived from the National Family Health Survey (NFHS-5) data used for model analysis.



Thank you!